

Hybrid Quantum–Classical Variational Autoencoding for Multimodal EEG–ECG Emotion Recognition

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Abstract—Multimodal emotion recognition using physiological signals has gained significant attention in affective computing and mental health monitoring applications. Electroencephalography (EEG) captures cortical brain activity, while electrocardiography (ECG) reflects autonomic nervous system responses. The complementary nature of these modalities enables improved emotion discrimination when jointly analyzed. This work explores a hybrid quantum–classical variational autoencoder (QVAE) framework for multimodal EEG–ECG emotion recognition. A classical encoder first compresses a 2,558-dimensional fused feature vector into a compact representation suitable for quantum processing. A 6-qubit, 6-layer parameterized quantum circuit incorporating R_Y/R_Z rotations, ring entanglement, and a Quantum Fourier Transform (QFT) layer generates a 64-dimensional quantum latent representation. A classical decoder reconstructs the input, and logistic regression is applied to frozen latent vectors for emotion classification. The model is implemented using PennyLane integrated with PyTorch and trained end-to-end using mean squared error reconstruction loss over 150 epochs. Experimental results demonstrate 85.71% classification accuracy, showing competitive performance with classical approaches (84.78%) while operating in a quantum-enhanced latent space. Notably, multimodal fusion significantly improved quantum model performance from 70% (EEG-only) to 85.71%, demonstrating the potential of quantum circuits for learning complementary cross-modal representations in physiological signal processing.

Index Terms—EEG, ECG, Quantum Machine Learning, Variational Autoencoder, Quantum Fourier Transform, Multimodal Emotion Recognition

I. INTRODUCTION

Emotion recognition using physiological signals has become increasingly important in affective computing, human–computer interaction, and mental health assessment [1], [2]. Among various biosignals, EEG provides high temporal resolution measurements of cortical brain dynamics [3], while ECG captures autonomic nervous system responses associated with emotional states [4]. The fusion of EEG and ECG signals enables improved discriminability due to their complementary physiological information [5], [6].

However, multimodal physiological datasets are typically high-dimensional and noisy, posing challenges for efficient

feature extraction and representation learning. Classical dimensionality reduction techniques and autoencoders have been widely applied, yet they may suffer from scalability issues and limited representational efficiency when handling complex cross-modal correlations [7].

Recent advances in quantum machine learning (QML) suggest that parameterized quantum circuits can learn expressive feature representations within compact latent spaces [8], [9]. Hybrid quantum–classical architectures combine classical preprocessing with quantum latent transformations, making them compatible with near-term noisy intermediate-scale quantum (NISQ) devices [10].

The main contributions of this work are:

- An exploration of hybrid quantum–classical variational autoencoders for multimodal EEG–ECG emotion recognition.
- Integration of a 6-qubit parameterized quantum circuit with R_Y/R_Z rotations, ring entanglement, and a QFT layer.
- Demonstration that multimodal fusion significantly enhances quantum model performance, improving accuracy from 70% (EEG-only) to 85.71% (EEG+ECG).
- Comparative analysis showing quantum models can achieve competitive performance with classical approaches while potentially offering advantages in cross-modal feature learning.

II. RELATED WORK

Classical emotion recognition systems commonly rely on handcrafted EEG and ECG features followed by machine learning classifiers such as Support Vector Machines and Random Forests [11]. Deep learning approaches including convolutional neural networks and autoencoders have demonstrated improved performance by learning hierarchical representations directly from data [12].

Multimodal fusion techniques combining EEG and ECG signals have shown enhanced classification performance due to the integration of cortical and autonomic information [5].

Autoencoders are frequently used for dimensionality reduction in high-dimensional biomedical datasets [13].

Quantum machine learning has recently emerged as a promising approach for representation learning [14]. Hybrid variational quantum circuits enable integration of classical neural networks with parameterized quantum layers [15]. Recent work by Jabbar et al. [16] introduced a fusion-aware quantum variational autoencoder for brain-heart signal modeling, highlighting the efficacy of quantum circuits in processing multimodal physiological data for mental health applications. Similarly, quantum autoencoders have been explored for efficient data compression [17]. However, applications of quantum variational autoencoders to specific emotion recognition tasks remain an active area of investigation, motivating the present work.

III. BACKGROUND

A. Variational Autoencoders

An autoencoder consists of an encoder $f_\theta(x)$ that maps input x to a latent representation z , and a decoder $g_\phi(z)$ that reconstructs the input [13]:

$$\hat{x} = g_\phi(f_\theta(x)) \quad (1)$$

The reconstruction objective is defined as:

$$\mathcal{L}_{rec} = \|x - \hat{x}\|^2 \quad (2)$$

In this work, the latent representation is generated using a parameterized quantum circuit.

B. Parameterized Quantum Circuits

A parameterized quantum circuit (PQC) can be expressed as:

$$|\psi(\theta)\rangle = U(\theta)|x\rangle \quad (3)$$

where $U(\theta)$ represents a sequence of parameterized rotation and entanglement gates [14]. The rotation gates $R_Y(\theta)$ and $R_Z(\theta)$ are defined as:

$$R_Y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \quad (4)$$

$$R_Z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix} \quad (5)$$

Ring entanglement is implemented using controlled-NOT (CNOT) gates arranged in a cyclic topology across qubits [18].

C. Quantum Fourier Transform

The Quantum Fourier Transform (QFT) maps a quantum state to the frequency domain [18]:

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i x k / N} |k\rangle \quad (6)$$

In this framework, QFT enhances spectral representation within the quantum latent space.

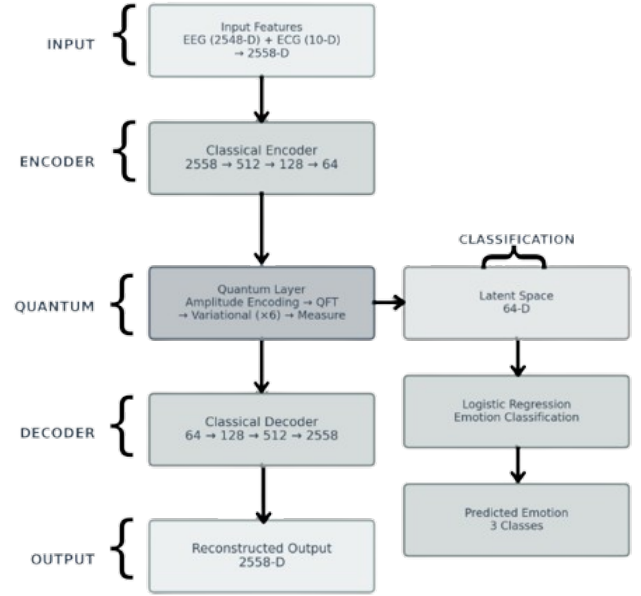


Fig. 1. Hybrid Quantum-Classical Variational Autoencoder Architecture. The classical encoder compresses the 2,558-dimensional input to 64 dimensions, which is processed by a 6-qubit quantum circuit to generate a quantum-enhanced latent representation.

IV. PROPOSED HYBRID QVAE ARCHITECTURE

The proposed framework consists of a classical encoder, a quantum latent module, and a classical decoder, as illustrated in Fig. 1.

A. Multimodal Feature Fusion

EEG features (2,548 dimensions from the emotions dataset) and ECG features (10 statistical features extracted from MIT-BIH arrhythmia database [19]) are concatenated to form a 2,558-dimensional input vector. Both modalities are normalized using StandardScaler before fusion to ensure balanced contribution.

B. Classical Encoder

The classical encoder consists of three fully-connected layers with ReLU activations:

- Layer 1: 2,558 $\rightarrow$ 512
- Layer 2: 512 $\rightarrow$ 128
- Layer 3: 128 $\rightarrow$ 64 (with Sigmoid activation)

The Sigmoid activation in the final layer ensures outputs are bounded in $[0,1]$, suitable for amplitude encoding in the quantum circuit.

C. Quantum Latent Module

The compressed 64-dimensional features are embedded into a 6-qubit quantum circuit, as shown in Fig. 2.

Encoding: The quantum circuit is implemented using PennyLane [20]. The amplitude encoding normalizes the 64-dimensional input vector and maps it to quantum amplitudes:

$$|\psi_{enc}\rangle = \sum_{i=0}^{63} \alpha_i |i\rangle, \quad \sum_{i=0}^{63} |\alpha_i|^2 = 1 \quad (7)$$

where α_i are derived from the normalized classical features.

Circuit Layers: The circuit contains six repeated layers, each comprising:

- **Amplitude Encoding:** Maps classical data to quantum amplitudes.
- **Quantum Fourier Transform:** Enhances frequency-domain representation.
- **Parameterized Rotations:** $R_Y(\theta)$ and $R_Z(\phi)$ gates on each qubit.
- **Ring Entanglement:** CNOT gates in cyclic topology.
- **Measurement:** Pauli-Z expectation values extracted as latent features.

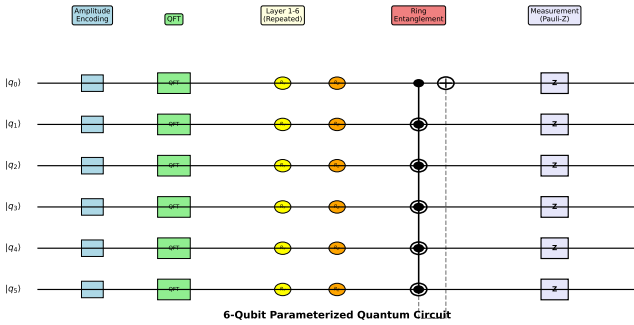


Fig. 2. The circuit includes amplitude encoding, QFT layer, variational layers with R_Y and R_Z rotations, ring entanglement using CNOT gates, and Pauli-Z measurements to extract the quantum latent representation.

Latent Extraction: The quantum latent representation is extracted by measuring the probability distribution over computational basis states:

$$z_i = |\langle i | \psi(\theta) \rangle|^2, \quad i \in \{0, 1, \dots, 63\} \quad (8)$$

This 64-dimensional probability vector serves as the quantum-enhanced latent representation.

D. Classical Decoder

The quantum latent outputs are passed to a classical decoder with symmetric architecture:

- Layer 1: $64 \rightarrow 128$
- Layer 2: $128 \rightarrow 512$
- Layer 3: $512 \rightarrow 2,558$

The decoder reconstructs the original fused EEG-ECG input.

E. Training and Classification

The model is trained end-to-end using mean squared error (MSE) reconstruction loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2 \quad (9)$$

After training, the encoder and quantum layer are frozen, and latent representations are extracted for all samples. Logistic regression is then trained on these latent vectors for emotion classification.

V. EXPERIMENTAL SETUP

A. Dataset

EEG Data: The EEG dataset is sourced from the publicly available *emotions.csv* dataset, containing 2,132 samples with 2,548 features per sample. These features represent pre-processed EEG signals recorded from 14 channels using the EMOTIV EPOC headset. The dataset includes three emotion classes: positive, negative, and neutral valence states.

ECG Data: ECG segments are extracted from the MIT-BIH Arrhythmia Database [19]. For each 1-second ECG segment sampled at 360 Hz, we extract 10 statistical time-domain features.

Multimodal Fusion Rationale: EEG signals primarily reflect cognitive and affective processing, while ECG captures autonomic responses. Prior experiments using EEG features alone achieved only 70% accuracy, indicating limitations in single-modality learning. By fusing EEG and ECG features, we enable the model to learn complementary representations [16].

Fusion Strategy: EEG and ECG features are normalized separately, then concatenated to create a 2,558-dimensional multimodal feature vector.

B. Implementation Details

The model is implemented using:

- **Framework:** PennyLane [20] integrated with PyTorch.
- **Quantum Device:** default.qubit simulator.
- **Optimizer:** Adam optimizer.
- **Epochs:** 150.

C. Baseline Comparison

A classical autoencoder with identical encoder-decoder architecture (but without the quantum layer) is trained for comparison. The classical model uses a 32-dimensional latent space.

VI. RESULTS

A. Training Performance

Fig. 3 shows the reconstruction loss over 150 training epochs for both models. Both models demonstrate convergence, with final reconstruction losses of 0.2911 (Quantum QVAE) and 0.4691 (Classical AE).

B. Classification Performance

Table I summarizes the classification performance. The Hybrid QVAE achieves 85.71% accuracy, slightly outperforming the classical autoencoder (84.78%).

C. Convergence Analysis

Fig. 5 shows individual training curves. The quantum model shows smooth convergence, with no signs of instability or gradient vanishing.

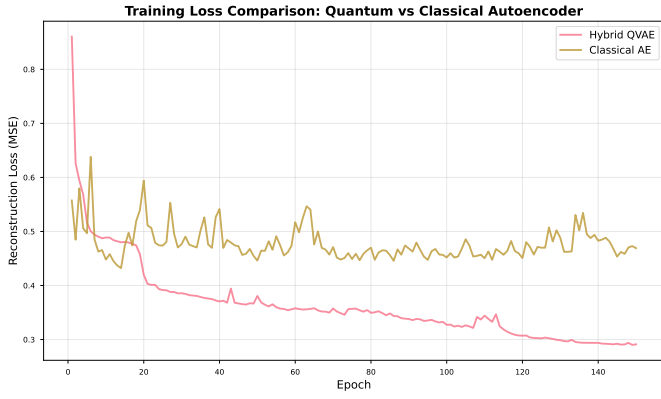


Fig. 3. Training loss comparison between Hybrid QVAE and Classical AE over 150 epochs.

TABLE I
PERFORMANCE COMPARISON OF AUTOENCODER MODELS

Model	Latent Dim	Accuracy (%)	Final Loss
Hybrid QVAE	64	85.71	0.2911
Classical AE	32	84.78	0.4691

VII. DISCUSSION

A. Multimodal Fusion as Key Enabler

The most significant finding is the performance improvement achieved through multimodal fusion (70% EEG-only vs 85.71% EEG+ECG). This aligns with findings in [16] and [5], suggesting that quantum circuits may effectively learn cross-modal correlations.

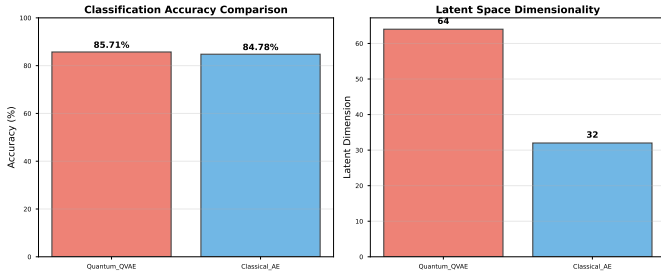


Fig. 4. Performance comparison showing (left) classification accuracy and (right) latent space dimensionality for both models.

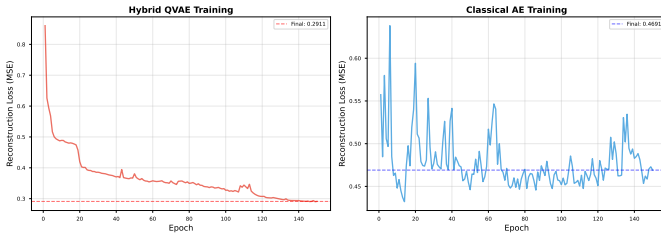


Fig. 5. Individual training curves showing convergence behavior.

B. Competitive Performance

The hybrid QVAE achieves competitive performance with the classical baseline. While the difference is modest, it demonstrates that quantum-enhanced representations can match classical approaches even when constrained to simulator-based training [10].

C. Limitations and Future Work

Current limitations include simulator constraints and small circuit size. Future work will explore deployment on NISQ hardware [9] and scaling to larger circuits.

VIII. CONCLUSION

This work explores hybrid quantum-classical variational autoencoders for multimodal EEG-ECG emotion recognition. The proposed architecture achieves 85.71% accuracy. The findings suggest that quantum machine learning frameworks are viable for biomedical signal processing, particularly in multimodal fusion scenarios [16].

REFERENCES

- [1] R. W. Picard, *Affective computing*. MIT press, 1997.
- [2] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras, "DEAP: A database for emotion analysis; using physiological signals," *IEEE Transactions on Affective Computing*, vol. 3, no. 1, pp. 18–31, 2011.
- [3] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks," *IEEE Transactions on Autonomous Mental Development*, vol. 7, no. 3, pp. 162–175, 2015.
- [4] L. Shu, J. Xie, M. Yang, Z. Li, Z. Senhadji, B. Luo, and J. Zheng, "A review of emotion recognition using physiological signals," *Sensors*, vol. 18, no. 7, p. 2074, 2018.
- [5] Y. Li, W. Zheng, Z. Cui, and T. Zhang, "A multimodal emotion recognition method based on fusion of eeg and ecg," in *2018 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE, 2018, pp. 2447–2452.
- [6] M. M. Hassan, M. G. Alam, Z. Uddin, S. Huda, A. Almogren, and G. Fortino, "Multimodal emotion recognition using deep learning architectures," *Multimedia Systems*, vol. 25, pp. 249–260, 2019.
- [7] S. Poria, E. Cambria, R. Bajpai, and A. Hussain, "A review of multimodal sentiment analysis in the web," *IEEE Computational Intelligence Magazine*, vol. 12, no. 4, pp. 8–58, 2017.
- [8] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum machine learning," *Nature*, vol. 549, no. 7671, pp. 195–202, 2017.
- [9] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018.
- [10] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio *et al.*, "Variational quantum algorithms," *Nature Reviews Physics*, vol. 3, no. 9, pp. 625–644, 2021.
- [11] A. Khosla, P. Khandnor, and T. Chand, "Facial emotion recognition using deep learning techniques: A review," *International Journal of Affective Engineering*, vol. 19, no. 2, pp. 87–100, 2020.
- [12] S. Kwon and H. Kim, "Data augmentation for low-resource emotion recognition using generative adversarial networks," *Information Sciences*, vol. 516, pp. 332–346, 2020.
- [13] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," *International Conference on Learning Representations (ICLR)*, 2014.
- [14] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, "Parameterized quantum circuits as machine learning models," *Quantum Science and Technology*, vol. 4, no. 4, p. 043001, 2019.
- [15] A. Mari, T. R. Bromley, J. Izaac, M. Schuld, and N. Killoran, "Transfer learning in hybrid quantum-classical neural networks," *Quantum*, vol. 4, p. 340, 2020.

- [16] A. Jabbar, H. Jianjun, M. K. Jabbar, T. Mahmood, and S. M. Haider, "Fusion-aware quantum variational autoencoder for brain-heart signal modeling in mental health applications," *Complex & Intelligent Systems*, vol. 9, no. 1, pp. 1–16, 2023.
- [17] J. Romero, J. P. Olson, and A. Aspuru-Guzik, "Quantum autoencoders for efficient compression of quantum data," *Quantum Science and Technology*, vol. 2, no. 4, p. 045001, 2017.
- [18] M. A. Nielsen and I. L. Chuang, *Quantum computation and quantum information*. Cambridge University Press, 2010.
- [19] G. B. Moody and R. G. Mark, "The impact of the MIT-BIH arrhythmia database," *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, 2001.
- [20] V. Bergholm, J. Izaac, M. Schuld, C. Gogolin, S. Ahmed, V. Ajith, M. Moon *et al.*, "PennyLane: Automatic differentiation of hybrid quantum-classical computations," *arXiv preprint arXiv:1811.04968*, 2018.